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Split valve air curtain

Abstract

Contamination from outgassing during a deposition process is addressed by a series of equipment enhancements, including throttle valves, a dual air curtain, and a residual gas analysis (RGA) monitor. The dual air curtain can be configured to flow a first gas during wafer processing and a second gas during wafer unloading, to re-direct and capture outgassed species. The dual air curtain and the throttle valves can be programmed in an automated feedback control system that utilizes data from the RGA monitor.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This patent application claims benefit of U.S. Provisional Patent Application No. 63/277,037 filed on Nov. 8, 2021 and titled “Slit Valve Air Curtain,” which is incorporated by reference herein in its entirety.

BACKGROUND
(1) With advances in semiconductor technology, there has been increasing demand for higher storage capacity, faster processing systems, higher performance, and lower costs. To meet these demands, the semiconductor industry continues to scale down the dimensions of semiconductor devices, such as metal oxide semiconductor field effect transistors (MOSFETs), including planar MOSFETs and fin field effect transistors (FinFETs). Such scaling down has increased the complexity of semiconductor manufacturing processes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with common practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1**A is a cross-sectional view of a via on a semiconductor wafer, in accordance with some embodiments.
(3) FIG. **1**B is an example surface chemical reaction for creating layers of the via shown in FIG. **1**A, in accordance with some embodiments.
(4) FIG. **2** is an illustrated flow diagram of a method for forming the via shown in FIG. **1**A, in accordance with some embodiments.
(5) FIG. **3** is a side elevation view of an interior of a processing chamber, in accordance with some embodiments.
(6) FIG. **4**A is a top plan view of an interior of a multi-chamber deposition equipment set, in accordance with some embodiments.
(7) FIG. **4**B is a series of cross-sectional views illustrating a process that occurs in the equipment set shown in FIG. **4**A, in accordance with some embodiments.
(8) FIG. **5** is a side elevation view of an interior of a processing chamber with a dual air curtain, in accordance with some embodiments.
(9) FIG. **6**A is a side elevation view of the processing chamber of FIG. **5** during processing and while a slit valve is closed, in accordance with some embodiments.
(10) FIG. **6**B is a side elevation view of the processing chamber of FIG. **5** during wafer unloading and while a slit valve is open, in accordance with some embodiments.
(11) FIG. **7** is a simulated pressure map associated with a dual air curtain, in accordance with some

embodiments.

(12) FIG. **8** is a top plan view of an interior of a multi-chamber deposition equipment set with hardware enhancements for reducing contamination, in accordance with some embodiments.

(13) FIG. **9** is a flow diagram of a method for implementing an automatic feedback control system to reduce contamination of a multi-chamber deposition equipment set, in accordance with some embodiments.

(14) FIG. **10** is a block diagram of a computer system that can serve to implement various embodiments of the present disclosure.

DETAILED DESCRIPTION

(15) The following disclosure provides different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed that are between the first and second features, such that the first and second features are not in direct contact.

(16) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(17) The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values is typically due to slight variations in manufacturing processes or tolerances.

(18) In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of a target value (e.g., ±1%, ±2%, ±3%, ±4%, and ±5% of the target value).

(19) The term “vertical,” as used herein, means nominally perpendicular to the surface of a substrate.

(20) It is to be appreciated that the Detailed Description section, and not the Abstract of the Disclosure section, is intended to be used to interpret the claims. The Abstract of the Disclosure section may set forth one or more but not all possible embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the subjoined claims in any way.

(21) During deposition of films in a process chamber, volatile substances can outgas and coat surfaces of the process chamber with chemical residues. When the coated surfaces cover moving parts of the process chamber, there can be a higher risk of disturbing accumulated residues and causing residue defects on the exposed surfaces of wafers during the semiconductor manufacturing process. Moving parts of the process chamber can include, for example, pressure valves or flow valves that open and close during processing, and entry/exit doors through which wafers pass during load and unload procedures. Through the use of in-situ particle monitoring and preventive measures that avoid or reduce changes in pressure and flow, the probability of generating residue defects can be reduced or eliminated, according to some embodiments of the present disclosure. Such improvements have multiple benefits, by increasing time between equipment maintenance events, improving product quality and yield, and increasing output efficiency.

(22) FIG. **1**A illustrates an exemplary film stack of via layers **100** formed on a semiconductor wafer

as part of a metal interconnect structure, according to some embodiments. Via layers **100** can be deposited on a substrate **102**.

(23) Substrate **102** can be a bulk semiconductor wafer or the top semiconductor layer of a semiconductor-on-insulator (SOI) wafer (not shown), such as silicon-on-insulator. In some embodiments, substrate **102** can include a crystalline semiconductor layer with its top surface parallel to (100), (110), (111), or c-(0001) crystal plane. In some embodiments, substrate **102** can be a glass or plastic substrate. Substrate **102** can be made of a semiconductor material such as, but is not limited to, silicon (Si). In some embodiments, substrate **102** can include (i) an elementary semiconductor, such as germanium (Ge); (ii) a compound semiconductor including silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); (iii) an alloy semiconductor including silicon germanium carbide (SiGeC), silicon germanium (SiGe), gallium arsenic phosphide (GaAsP), gallium indium phosphide (InGaP), gallium indium arsenide (InGaAs), gallium indium arsenic phosphide (InGaAsP), aluminum indium arsenide (InAlAs), and/or aluminum gallium arsenide (AlGaAs); or (iv) a combination thereof. Further, substrate **102** can be doped with p-type dopants (e.g., boron (B), indium (In), aluminum (Al), or gallium (Ga)) or n-type dopants (e.g., phosphorus (P) or arsenic (As)). In some embodiments, different portions of substrate **102** can have opposite type dopants.

(24) In some embodiments, substrate **102** is a wafer, e.g., a semiconductor wafer, in which transistors or other electronic devices have been fabricated. A top layer of substrate **102** can be, for example, a contact layer that helps to form a low-resistivity electrical contact to underlying devices. Such a contact layer can be made of, for example, cobalt, nickel, or silicides thereof. In some embodiments, substrate **102** further includes, above the contact layer, a liner made of e.g., titanium, titanium nitride, or a combination thereof.

(25) Via layers **100** can include, for example, a seed layer **106**, and a bulk metal **108**, surrounded by a sidewall insulating material **110** and an inter-layer dielectric (ILD) **112**. In some embodiments, seed layer **106** and bulk metal **108** both include the same primary metal component, e.g., tungsten (W). Seed layer **106** can have a thickness between about 27 Å and about 33 Å. ILD **112** can include silicon dioxide (SiO.sub.2) or a low-k dielectric material such as, for example, a fluorosilicate glass, a carbon-doped silicon dioxide, a porous silicon dioxide, a porous carbon-doped silicon dioxide, a polyimide, or a polytetrafluoroethylene (PTFE). Via layers **100** are used to illustrate the present disclosure by way of example. However, the present disclosure is not so limited. Other film stacks may pose a similar defect risk as the via layers described herein.

(26) FIG. **1**B illustrates formation of seed layer **106**, according to some embodiments. Seed layer **106** can be deposited onto substrate **102** under vacuum by, for example, chemical vapor deposition (CVD) or plasma enhanced chemical vapor deposition (PECVD), by exposing substrate **102** to a gas mixture containing boron hexafluoride (B.sub.2H.sub.6) and tungsten hexafluoride (WF.sub.6) gases. When the fluorine component of WF.sub.6 dissociates and reacts chemically with B.sub.2H.sub.6, tungsten is deposited onto substrate **102**, and the fluorine, boron, and hydrogen components are released as volatile reaction products. Volatilized chemicals may then fill the vacuum chamber and coat interior walls during the reaction phase, before the chamber is evacuated. Seed layer **106** is used to illustrate the present disclosure by way of example. However, the present disclosure is not so limited. Deposition of other seed layers, or other types of layers, may pose a similar defect risk as seed layer **106** that is described herein.

(27) FIG. **2** shows an exemplary method **200** that can be executed in a multi-chamber deposition equipment set **220**, according to some embodiments. Method **200** can be used to form the conductive seed and bulk tungsten layers of the stack of via layers **100** shown in FIG. **1**A. A top-down view of multi-chamber deposition equipment set **220** is provided in FIG. **2** for each operation in method **200**. In some embodiments, multi-chamber deposition equipment set **220** includes a plurality of load stations **221** (for example, three load stations, **221***a*-**221***c*, are shown in FIG. **2**), an

entrance load lock **222**, a buffer chamber **223**, a plurality of process chambers **224** (for example, four process chambers, **224***a*-**224***d*, are shown in FIG. **2**), and an exit load lock **226**. Multi-chamber deposition equipment set **220** can further include one or more robots for automated transfer of wafers to and from the various stations and chambers, or modules. In some embodiments, buffer chamber **223** and process chambers **224** stay under vacuum, while load locks **222** and **226**, adjoining both buffer chamber **223** and load stations **221**, alternate between atmospheric pressure and vacuum. Process chambers **224** are plumbed with process gases to chemically alter the surface of the wafer being processed, while buffer chamber **223** is used for transferring wafers among process chambers **224**. In some embodiments, one or more process chambers **224** can be configured for CVD or for PECVD, with the addition of an energy source that can ionize process gases to create a plasma. In some embodiments, one or more process chambers **224** can be configured as a variant of CVD, such as atomic layer deposition (ALD) or spatial atomic layer deposition (SALD), in which surface reactions create films on the wafer in process, one atomic layer at a time. An exemplary equipment configuration for multi-chamber deposition equipment set **220** is used to illustrate the present disclosure by way of example. However, the present disclosure is not so limited. Other equipment configurations, or other equipment sets used for deposition and/or etching of films on wafers, can pose a similar defect risk as the examples described herein.
(28) In accordance with exemplary method **200**, an automated robot can be programmed to transfer a single wafer through multi-chamber deposition equipment set **220**. At operation **202**, the wafer can be transferred from a front opening unified pod (FOUP) positioned at load station **221***a* of multi-chamber deposition equipment set **220** to a seed layer deposition chamber **224***c* via entrance load lock **222** and buffer chamber **223**. Entrance load lock **222** can have two doors (not shown) so that a wafer enters entrance load lock **222** at atmospheric pressure, from load station **221***a* through a first door, then entrance load lock **222** is evacuated, and the wafer exits entrance load lock **222** under vacuum, into buffer chamber **223** through a second door. At operation **204**, seed layer **106** is deposited onto the wafer in seed layer deposition chamber **224***c*. At operation **206**, the wafer is transferred from seed layer deposition chamber **224***c* to bulk layer deposition chamber **224***d* via buffer chamber **223**. At operation **208**, bulk metal **108** is deposited onto the wafer in bulk layer deposition chamber **224***d*. At operation **210**, the wafer is transferred from bulk layer deposition chamber **224***d* via buffer chamber **223** and exit load lock **226**, back to the FOUP at load station **221***a* of multi-chamber deposition equipment set **220**. Exit load lock **226** can have two doors (not shown) so that the wafer enters exit load lock **226** from buffer chamber **223** under vacuum, through a first door, exit load lock **226** is pressurized, and the wafer is removed from exit load lock **222**, at atmospheric pressure, through a second door for loading back into the FOUP at load station **221***a*.
(29) Referring to FIG. **2**, in operation **204**, contaminants **300** are formed within seed layer deposition chamber **224***c* as shown in FIG. **3**. FIG. **3** illustrates an interior view of seed layer deposition chamber **224***c*, according to some embodiments. Seed layer deposition chamber **224***c* can have components such as a pump **302**, a heated chuck **304**, a showerhead **306**, and a slit valve **308**. Slit valve **308** serves as a doorway for loading and unloading wafers onto heated chuck **304** for processing. Gas phase reactants enter seed layer deposition chamber **224***c* through showerhead **306**, and react at the surface of substrate **102** to form seed layer **106**. Residual gases and reaction products exit seed layer deposition chamber **224***c* via pump **302**. In some embodiments, the temperature of heated chuck **304** is between about 270 degrees C. and about 330 degrees C. Heated chuck **304** may catalyze surface reactions, and/or increase a deposition rate of seed layer **106**. The temperature of heated chuck **304** can affect the smoothness of the surface of seed layer **106**. Heated chuck **304** may also cause evaporation or outgassing of chemicals from the surface of seed layer **106**, which chemicals can condense into contaminants **300**. In some embodiments, contaminants **300** are as large as about 10 nm in diameter. The composition of contaminants **300** can include tungsten, titanium, and nitrogen, with tungsten being the most prevalent species. Although pump **302** is intended to remove residual gases from seed layer deposition chamber **224***c*, contaminants

300 that are not captured by pump **302** can accumulate on slit valve **308**. After some period of time, when a thick enough layer forms on slit valve **308**, motion of slit valve **308** while opening and closing can dislodge contaminants **300**, creating a particle source that can impact wafers as they pass through slit valve **308**.

(30) Referring to FIG. **2**, in operations **204** and **206**, formation and migration of contaminants **300** can occur as shown in FIG. **4**A. FIG. **4**A shows an enlarged view of multi-chamber deposition equipment set **220** during operations **204** and **206**. FIG. **4**A further illustrates contaminants **300**, which may be generated during operation **204** as described above, migrating out of seed layer deposition chamber **224***c*, into buffer chamber **223**, during wafer transfer to bulk layer deposition chamber **224***d* in operation **206**. As the wafer moves out of seed layer deposition chamber **224***c* through slit valve **308**, in addition to residues being dislodged by motion of slit valve **308**, changes in gas flows and/or chamber pressure may further disturb residues and spread contaminants to other areas of multi-chamber deposition equipment set **220**. Such residues can impact wafers, causing contaminants **300** to accumulate on the top surface of seed layer **106**. Contaminants **300** may also be swept out from seed layer deposition chamber **224***c* as the wafer passes along a transfer path **400** into buffer chamber **223**. In addition, wafers may continue to outgas volatile reaction products from seed layer **106** while they are being transported through buffer chamber **223**, thus contaminating buffer chamber **223**.

(31) Referring still to FIG. **2**, in operations **204** and **206**, an effect of contaminants **300** at the interface between seed layer **106** and bulk metal **108** on a wafer is illustrated in FIG. **4**B. FIG. **4**B shows the wafer at three successive times during operations **204-206**. In a top frame of FIG. **4**B, at time t.sub.1, the wafer is shown after deposition of seed layer **106** has been completed, while the wafer is still in seed layer deposition chamber **224***c*. In a middle frame of FIG. **4**B, at time t.sub.2, contaminants **300** land on seed layer **106** as the wafer exits process chamber **224***c* through slit valve **308** and into buffer chamber **223**. In a bottom frame of FIG. **4**B, contaminants **300** remain on the surface of the wafer as it continues along transfer path **400**, into bulk layer deposition chamber **224***d* for processing at operation **206**. Bulk metal **108** is then deposited over contaminants **300**, which intervene between seed layer **106** and bulk metal **108**. The presence of contaminants **300** at the interface between the two tungsten layers will thus obstruct electrical contact at the interface and will introduce via resistance that can result in reduced device performance.

(32) FIG. **5** illustrates an interior view of seed layer deposition chamber **224***c* shown in FIG. **3**, following implementation of a dual air curtain **500**, according to some embodiments. Dual air curtain **500** can be disposed between slit valve **308** and pump **302**, so that wafers pass through dual air curtain **500** while loading and unloading from seed layer deposition chamber **224***c*. When it is positioned above pump **302**, dual air curtain **500** produces a region of laminar flow that separates the processing area of seed layer deposition chamber **224***c* between heated chuck **304** and showerhead **306** from the wafer transfer area near slot valve **308**. Dual air curtain **500** can be designed to flow first and second inert gases, **502** and **504**, respectively at flow rates in the range of about 500 standard cubic centimeters per minute (sccm) to about 1000 sccm. In some embodiments, nitrogen gas (N.sub.2) is used as first inert gas **502** and argon (Ar) us used as second inert gas **504**. Alternatively, other inert gases, e.g., helium (He) or oxygen (O.sub.2), can be used. In some embodiments, the same gas can be used as both first and second inert gases **502** and **504**.

(33) Dual air curtain **500** can alter the motion of contaminants **300** by directing outgassed reaction products toward pump **302** for removal from seed layer deposition chamber **224***c*. Dual air curtain **500** therefore can prevent contaminants **300** from accumulating on slit valve **308**. When only a thin layer of contaminants **300** accumulates on slit valve **308**, the thin layer is more likely to stay intact. Thus, the likelihood of flaking and causing defects on processed wafers is lowered. In addition, the presence of dual air curtain **500** increases pressure inside seed layer deposition chamber **224***c*, which allows increasing pressure in the buffer chamber to balance the chamber pressure and prevent contaminants from rushing out of seed layer deposition chamber **224***c* during wafer

transfer. In some embodiments, the buffer chamber pressure can be set to about 250 mTorr.
(34) Referring to FIGS. **6**A and **6**B, dual air curtain **500** can be implemented to flow different inert gases at different times, in accordance with some embodiments. FIG. **6**A shows an operation of dual air curtain **500** at time t.sub.A when a wafer is present on heated chuck **304** and is being processed in seed layer deposition chamber **224***c*, while slit valve **308** is closed. At time t.sub.A, dual air curtain **500** can be programmed to initiate flow of second inert gas **504**, e.g., Ar, to direct reaction byproducts and contaminants **300** toward pump **302**. As chemicals outgas from the surface of seed layer **106**, dual air curtain **500** blocks motion of volatilized species from contaminating slit valve **308** or migrating to buffer chamber **123**.
(35) FIG. **6**B shows an operation of dual air curtain **500** at time t.sub.B when a wafer is being unloaded from seed layer deposition chamber **224***c* and slit valve **308** is open. At time t.sub.B, dual air curtain **500** can be programmed to turn off the flow of second gas **504** and turn on the flow of first inert gas **502**, e.g., N.sub.2, to direct reaction byproducts and contaminants **300** toward pump **302**. As the wafer is transported out of the chamber towards open slit valve **308**, the flow of inert gas **502** through dual air curtain **500** sweeps volatilized species from the surface of seed layer **106**, confining contaminants **300** to the chamber, where they can be removed by pump **302**.
(36) In some embodiments, different gases can be plumbed to dual air curtain **500**, and flow control can be tailored to specific processes other than the example of depositing seed layer **106**. For example, dual air curtain **500** can be programmed to flow any combination of first and second inert gases at various times, as needed for the process that is being used in the chamber. In some embodiments, dual air curtain **500** can be implemented on other types of semiconductor process equipment with etching chambers instead of, or in addition to, deposition chambers. In some embodiments, dual air curtain **500** can be implemented on other platforms that may not include vacuum chambers, such as a solvent station for a lithography track where outgassing of volatile solvent chemicals can pose a similar problem to the outgassing described herein.
(37) FIG. **7** shows a magnified view of dual air curtain **500** in operation, with first inert gas **502** providing laminar flow. FIG. **7** corresponds to region indicated in FIG. **6**B by a dotted line box **700**, in which dual air curtain **500** is implemented on seed layer deposition chamber **224***c*, as described above with respect to FIG. **6**B. In FIG. **7**, the wafer is not in transit, allowing for laminar flow to proceed unobstructed. FIG. **7** shows simulated pressure gradients in the vicinity of slit valve **308**, defined in FIG. **7** as slit valve channel **702**. FIG. **7** also shows simulated pressure gradients in a laminar flow region **704** between dual air curtain **500** and the inlet of pump **302**.
(38) In FIG. **7**, regions of lowest pressure are in laminar flow region **704** and to the right of laminar flow region **704**, where the pressure can be less than 100 mTorr. A medium pressure region exists where slit valve channel **702** joins laminar flow region **704**, on the left side of laminar flow region **704**, where the pressure can be in the range of about 100-150 mTorr. The highest pressures are in slit valve channel **702**, farthest away from pump **302**, where pressure values can be greater than about 200 mTorr.
(39) FIG. **8** shows the multi-chamber deposition equipment set **220** of FIG. **4**A, implemented with additional hardware solutions to further reduce contamination from outgassing due to deposition of seed layer **106**, according to some embodiments. Additional hardware solutions shown in FIG. **8** include a chamber throttle valve **800** for improving control of pump **302**, a buffer chamber throttle valve **802** for regulating pressure in buffer chamber **223**, and a residual gas analyzer (RGA) monitor **804** installed in buffer chamber **223**. Additionally or alternatively, RGA monitor **804** can be installed in other chambers of multi-chamber deposition equipment set **220**. RGA monitor **804** measures gas composition of contaminants **300** using mass spectrometry. When installed in buffer chamber **223**, RGA monitor **804** can act as an in-situ process monitor to help determine, from the composition of detected contaminants, which chamber of multi-chamber deposition equipment set **220** is the source of detected contaminants.
(40) Throttle valves **800** and **802** can be configured separately, or in a coordinated fashion, to

regulate a differential pressure between buffer chamber **223** and a chamber of multi-chamber deposition equipment set **220** with slit valve **308**, e.g., seed deposition chamber **224***c*. With improved pressure regulation provided by throttle valves **800** and **802**, changes in pressure can be avoided, reducing the probability of disturbing residue buildup on slit valve **308**. In addition, the action of pump **302** and a separate buffer chamber pump (not shown) can be varied using throttle valves **800** and **802**, respectively, according to process needs, to more quickly evacuate volatilized species, thereby preventing residues from forming on equipment surfaces.

(41) RGA monitor **804** and throttle valves **800** and **802** can be implemented together as a feedback control system to reduce effects of contaminants **300**, and to reduce contamination of buffer chamber **223** and bulk deposition chamber **224***d*. Using RGA monitor **804** in-situ, while wafers are in process and in transit, throttle valves **800** and **802** can be adjusted in accordance with detected contaminant levels. Operation of dual air curtain **500** can also be implemented within the feedback control system

(42) FIG. **9** illustrates an exemplary method **900** for providing real-time feedback control of hardware solutions shown in FIG. **8**, during execution of steps **204** and **206** within method **200**, in accordance with some embodiments. Method **900** uses data from multiple types of monitors, in combination, for controlling multiple hardware elements installed in multi-chamber deposition equipment set **220** to reduce contamination from outgassing.

(43) At operation **902**, an in-situ RGA measurement is performed to detect contaminants in buffer chamber **123**.

(44) At operation **904**, a level of contaminants, e.g., contaminants having tungsten composition, determined in operation **902** is compared against a standard for failure data collection (FDC) to determine whether or not the contaminant level is within a maximum allowed limit. When the contaminant level is below a threshold value, method **200** continues and subsequent RGA measurements continue to be performed at operation **902** at prescribed time intervals.

(45) At operation **906**, when the level of contaminants determined in operation **902** exceeds the threshold value, which may indicate, for example, a burst of contaminants following opening of slit valve **308**, the status of an equipment particle monitor is checked using a tool automation program (TAP) to see if there is a commensurate increase in particle trend data.

(46) Equipment particle data may be stored on a tool server and may be used for statistical process control (SPC) of multi-chamber deposition equipment set **220**. In some embodiments, SPC refers to collecting equipment particle data on test wafers and/or collecting in-line defect data on product wafers, at regular time intervals, and monitoring particle trends in real time. Using SPC, outliers in the data can be recognized to control the process. Automated statistical control, using 3-sigma or 6-sigma (double-sided) threshold values, flags anomalies in particle data trends so that action can be taken to contain product and to address equipment failures. SPC can be used to monitor either equipment sets, individual chambers, or both.

(47) At operation **908**, numerical values of the RGA and equipment monitors can be combined to determine a composite particle score. Based on the composite particle score and/or a relative contribution of various particle monitor values to the overall particle score, equipment adjustment values can be determined and, if needed, can be codified in a recipe modification summary (RMS). For example, when the composite score exceeds a predetermined threshold, further investigation can be carried out to determine whether the largest contribution is from the RGA monitor in buffer chamber **223** or from specific automated tool monitors of individual process chambers **224**. If the contribution can be narrowed down to a specific chamber, e.g., process chamber **224***c*, maintenance can be done on the chamber to confirm the particle signature and eliminate the root cause of particles, for example, slit valve accumulation and flaking. The response to an out-of-control particle monitor may be to replace a contaminated slit valve **308**. On the other hand, if the RGA monitor is the larger contributor, the root cause may be narrowed down to facilities contamination, e.g., gas lines or pumps, or to contaminants coming from the product itself, e.g., outgassing. The

response to an out-of-control RGA monitor may be to adjust the operation of dual air curtain **500**. One advantage of using the RGA monitor and/or the RMS monitor is that the RGA monitor operates continuously, whereas SPC data is intermittent or periodic. So, the RGA monitor may flag a problem earlier than a periodic equipment particle monitor would.

(48) At operation **910**, adjustments to hardware elements are made according to the determined equipment adjustment values. Based on one or more of an RGA value, a TAP value, and an RMS value, relative flow rates of first and second inert gases **502** and **504** of dual air curtain **500** can be modified. For example, in response to an RGA value indicating an increase in contaminants **300** within buffer chamber **223**, the flow of first inert gas **502** through dual air curtain **500** can be increased. This would ensure that when wafers are transferred from process chamber **224***c*, outgassed contaminants **300** do not enter buffer chamber **223**. In response to a TAP value indicating an increase in particles within process chamber **224**, the flow of second inert gas **504** through dual air curtain **500** can be increased. This would ensure that while wafers are being processed in chamber **224***c*, outgassed contaminants **300** are swept into the draft of pump **302** before they can accumulate on slit valve **308**.

(49) Additionally or alternatively, settings for chamber throttle valve **800** can be adjusted, or settings for buffer chamber throttle valve **802** can be adjusted based on one or more of the RGA value, TAP value, and RMS value. For example, in response to an equipment particle monitor or TAP value increasing beyond a threshold value, throttle valves **800** and/or **802** can be opened so as to evacuate the chambers more quickly, thereby reducing the particulate load.

(50) Method **900** can be implemented with any combination of monitors contributing to the RMS calculation in operation **908**, and any combination of adjustments made at operation **910**, in response to the monitor data. Any or all of the equipment adjustments made at operation **910** can be made in real time, in response to real-time measurements. Thus, processing of wafers can be adjusted based on varying contamination levels, as they occur, without sacrificing yield.

(51) FIG. **10** is an illustration of an example computer system **1000** in which various embodiments of the present disclosure can be implemented. Computer system **1000** can be any well-known computer capable of performing the functions and operations described herein. Computer system **1000** can be used, for example, to execute one or more operations in method **200** of FIG. **2** and method **900** of FIG. **9**.

(52) Computer system **1000** includes one or more processors (also called central processing units, or CPUs), such as a processor **1004**. Processor **1004** is connected to a communication infrastructure or bus **1006**. Computer system **1000** also includes input/output device(s) **1003**, such as monitors, keyboards, pointing devices, etc., that communicate with communication infrastructure or bus **1006** through input/output interface(s) **1002**. Computer system **1000** also includes a main or primary memory **1008**, such as random access memory (RAM). Main memory **1008** can include one or more levels of cache. Main memory **1008** has stored therein control logic (e.g., computer software) and/or data. In some embodiments, the control logic (e.g., computer software) and/or data can include one or more of the operations described above with respect to method **200** of FIG. **2** and method **900** of FIG. **9**.

(53) Computer system **1000** can also include one or more secondary storage devices or memory **610**. Secondary memory **1010** can include, for example, a hard disk drive **1012** and/or a removable storage device or drive **1014**. Removable storage drive **1014** can be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

(54) Removable storage drive **1014** can interact with a removable storage unit **1018**. Removable storage unit **1018** includes a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **1018** can be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/any other computer data storage device. Removable storage drive **1014** reads from and/or writes to removable storage unit **1018** in a

well-known manner.
(55) According to some embodiments, secondary memory **1010** can include other means, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **1000**. Such means, instrumentalities or other approaches can include, for example, a removable storage unit **1022** and an interface **1020**. Examples of the removable storage unit **1022** and the interface **1020** can include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface. In some embodiments, secondary memory **1010**, removable storage unit **1018**, and/or removable storage unit **1022** can include one or more of the operations described above with respect to method **900** of FIG. **9**.
(56) Computer system **1000** can further include a communication or network interface **1024**. Communication interface **1024** enables computer system **1000** to communicate and interact with any combination of remote devices, remote networks, remote entities, etc. (individually and collectively referenced by reference number **1028**). For example, communication interface **1024** can allow computer system **1000** to communicate with remote devices **1028** over communications path **1026**, which can be wired and/or wireless, and which can include any combination of LANs, WANs, the Internet, etc. Control logic and/or data can be transmitted to and from computer system **1000** via communication path **1026**.
(57) The operations in the preceding embodiments can be implemented in a wide variety of configurations and architectures. Therefore, some or all of the operations in the preceding embodiments—e.g., method **200** of FIG. **2** and method **900** of FIG. **9**—can be performed in hardware, in software or both. In some embodiments, a tangible apparatus or article of manufacture comprising a tangible computer useable or readable medium having control logic (software) stored thereon is also referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **1000**, main memory **1008**, secondary memory **1010** and removable storage units **1018** and **1022**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **1000**), causes such data processing devices to operate as described herein.
(58) Semiconductor processing equipment can be configured with specialty hardware to reduce the effects of contamination that arises from the process itself, for example, from outgassing following film deposition. Such hardware can include throttle valves for regulating pump speed and efficiency, and air curtains such as a dual air curtain installed at the entrance to a process chamber. The dual air curtain can be programmable, and each of the hardware items can be controlled automatically in real time using in-situ monitoring of the process, combined with data from periodic equipment monitors that is stored on a server. A feedback control system can be used to make equipment adjustments that are tailored for certain processes, as needed.
(59) In some embodiments, a method includes: loading a wafer into a buffer chamber; transferring the wafer from the buffer chamber to a process chamber including a slit valve, a pump, and a dual air curtain disposed between the slit valve and the pump; processing the wafer in the process chamber; and transferring the wafer from the process chamber to the buffer chamber.
(60) In some embodiments, a system includes: a wafer station; a load lock adjoining the wafer station, the load lock configured to alternate between atmospheric pressure and vacuum; a buffer chamber adjoining the load lock; a vacuum chamber adjoining the buffer chamber and including a showerhead and a slit valve, the vacuum chamber configured to expose a wafer to a vaporized chemical from the showerhead; and a dual air curtain between the slit valve and the showerhead and configured to flow a plurality of gases.
(61) In some embodiments, a method includes: detecting a level of contaminants in a buffer

chamber using an in-situ residual gas analyzer (RGA) monitor; collecting data from a tool server; analyzing the level of contaminants together with the data from the tool server to determine one or more adjustment values; and in response to the level of contaminants being above a first threshold, adjust relative flow rates of different inert gases flowing through a dual air curtain based on the one or more adjustment values.

(62) The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art will appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art will also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A system, comprising: a wafer station; a load lock adjoining the wafer station, the load lock configured to alternate between atmospheric pressure and vacuum; a buffer chamber adjoining the load lock; a vacuum chamber adjoining the buffer chamber and comprising a showerhead and a slit valve, the vacuum chamber configured to expose a wafer to gas phase reactants from the showerhead; and a dual air curtain between the slit valve and the showerhead and configured to flow a plurality of gases.

2. The system of claim 1, wherein the vacuum chamber further comprises a chuck.

3. The system of claim 1, wherein the vacuum chamber is configured to perform one or more of a chemical vapor deposition (CVD) process, a plasma enhanced chemical vapor deposition (PECVD) process, an atomic layer deposition (ALD) process, and a spatial atomic layer deposition (SALD) process.

4. The system of claim 1, wherein the plurality of gases includes one or more of argon, nitrogen, oxygen, and helium.

5. The system of claim 1, further comprising a chamber throttle valve configured to regulate a pressure of the vacuum chamber.

6. The system of claim 1, further comprising a buffer chamber throttle valve configured to regulate a pressure of the buffer chamber.

7. The system of claim 1, further comprising a residual gas analysis (RGA) monitor configured to measure composition of residual gases in one or more of the vacuum chamber and the buffer chamber.

8. The system of claim 7, further comprising a controller configured to regulate one or more of gas pressure and gas flow based on measurements performed by the RGA monitor.

9. The system of claim 8, wherein the controller is configured to regulate the one or more of gas pressure and gas flow based on one or more of a failure data collection (FDC) monitor, data stored on a tool server, and a statistical process control (SPC) monitor.

10. The system of claim 8, wherein the controller is configured to control the dual air curtain to: flow a first gas while the wafer is exposed to the gas phase reactants in the vacuum chamber and the slit valve is closed; and flow a second gas while the wafer is being transferred out of the vacuum chamber and the slit valve is open.